

“MediAssist”

AI-Powered Emergency First Aid Guidance

When seconds matter, we guide you through it.

HACKATHON SUBMISSION · HEALTH TECH · AI FOR GOOD · 2026

400K+

Cardiac arrests yearly in the US

70%

Emergencies happen near bystanders

3×

Survival with timely bystander first aid

16 Emergency Scenarios · RAG Knowledge Base · Voice Mode · Offline Capable · Multilingual

1. Executive Summary

MediAssist is a real-time, AI-powered first aid guide. It walks you through any medical emergency step by step using guidance sourced directly from certified first aid handbooks. Most people freeze in emergencies. Not from cowardice but because they simply don't know what to do. MediAssist fixes that, instantly, on any phone.

Category	HealthTech / AI for Good
Tech Core	RAG + LLM + Voice Interface
Coverage	16 emergency types
Access	Web, Mobile

2. The Problem

The ambulance takes 20–30 minutes on average. Brain damage from cardiac arrest begins in 4. That gap is where MediAssist operates.

- Only 18% of people globally have received first aid training
- Bystanders are present at 70% of emergencies
- Panic + no knowledge = inaction. Inaction kills.
- Googling 'how to do CPR' during a crisis costs 3–5 minutes

What's Out There - And Why It Fails

Option	The Gap	MediAssist
Google Search	No structure, unfiltered, 3–5 min lost	Verified steps in under 5 seconds
First Aid Books	Not accessible in a panic	Conversational, guided, hands-free
Static First Aid Apps	No adaptive Q&A, no voice	Dynamic AI that asks the right questions
Emergency Hotlines	Busy lines, language barriers	Always on, multilingual, zero wait

3. The Numbers Don't Lie

These aren't abstract figures they're people who could be saved with the right guidance at the right moment.

17M Cardiac deaths/year globally	5.4M Snakebite incidents/year	11M Burns needing care yearly	54% Fewer ER visits with first aid
--	---	---	--

4 min	Brain damage begins in cardiac arrest. CPR doubles survival odds.
73%	Of choking deaths - the victim wasn't alone. A bystander could have helped.
3×	Higher survival rate when bystander first aid is given immediately.
80%	Of minor burns get worse when treated incorrectly (ice, butter, toothpaste).
\$1T+	Annual global economic cost of unintentional, largely preventable injuries.

4. MediAssist <How It Works>

MediAssist uses Retrieval-Augmented Generation (RAG) meaning every answer is pulled from certified first aid handbooks, not fabricated. Think of it as a paramedic in your pocket that only speaks from verified sources.

The Flow

01 DETECT	User describes the emergency in plain language. AI identifies the type instantly.
02 RETRIEVE	Relevant passages are pulled from indexed first aid handbooks (Red Cross, AHA, WHO).
03 RESPOND	AI generates clear, numbered, short instructions tailored to the situation.
04 ADAPT	System asks follow-up questions victim's age, consciousness, severity and adjusts.
05 ESCALATE	For serious cases, it prompts emergency services with a pre-filled situation summary.

Core Features

• RAG Accuracy Every step comes from a certified first aid book not the internet.	• Voice Mode Hands-free spoken guidance keep your hands on the patient.
• Works Offline No signal? No problem. Core protocols cached locally.	• Instant Launch One tap to start. No login, no menus just guidance.
• 12+ Languages Breaking language barriers in global emergencies.	• EMS Integration Auto-generates location + situation summary for dispatchers.

5. What MediAssist Covers

16 emergencies. Every one sourced from registered first aid textbooks. Every response tested against certified protocols.

Emergency	What MediAssist Guides You Through
Cardiac Arrest (CPR)	Compression depth/rate, AED use, rescue breathing
Choking	Back blows, Heimlich (adult + infant), unconscious protocol
Severe Bleeding	Tourniquet steps, wound packing, shock signs
Burns — Major	Stop-drop-roll, airway, burn area estimate

Asthma Attack	Inhaler technique, severe signs, EMS trigger
Snake Bite	Immobilize, no tourniquet myth, antivenom urgency
Electric Shock	Cut power first, AED readiness, entry/exit wounds
Heat Exhaustion / Stroke	Cooling technique, hydration, when it becomes critical
Head Injury	Concussion signs, AVPU check, spinal precautions
Seizures / Convulsions	Safety do's and don'ts, timing, post-ictal care
Diabetic Emergency	Hypo vs. hyper signs, sugar response, EMS triggers
Eye Injury	Rinse for chemicals, foreign body guide, no rubbing

6. Real-World Impact

What Changes When People Are Guided

Scenario	Outcome with Timely First Aid
Cardiac Arrest	Survival jumps from 10% → 30–50% with immediate CPR
Choking	Up to 65% of deaths preventable with Heimlich guidance
Minor Burns	Correct first aid reduces long-term tissue damage by 80%
Snake Bite	Avoiding tourniquet misuse saves 200K+ limbs yearly
Heat Stroke	Cooling within 10 min drops fatality risk from 17% to 3%
Seizures	Proper response prevents secondary injury in 90% of cases

Who It Serves

 Rural Communities Where ambulances take 20+ min and first aid is the only option.	 Schools & Workplaces Safety officers equipped with real-time AI guidance.	 Disaster Zones Offline mode keeps MediAssist working without any signal.
---	---	--

7. Real-World Cases - Saved & Lost

Every statistic in this proposal represents a real person. Below are recent documented incidents that illustrate both the power of first aid knowledge and the tragic cost of its absence.



A bystander assists someone in distress using first aid knowledge



Emergency medical personnel responding to a critical cardiac emergency

☒ LIVES SAVED

Bystanders with first aid knowledge have successfully performed the Heimlich manoeuvre, CPR, and wound control saving lives before ambulances arrived. An off-duty officer's training saved a 1-year-old girl in seconds.

☐ LIVES LOST

73% of choking deaths happen when a bystander is present but lacks knowledge. Panic, uncertainty, and the absence of step-by-step guidance turn preventable tragedies into fatalities. Inaction, not absence, is the killer.

Off-Duty Officer Saves Choking Toddler with Back Blows - Dec 2025

Source: KSLA / WABC News, December 28, 2025

A 1-year-old girl in New York was saved from choking when her mother rushed to a neighbour an off-duty NYPD officer who performed back blows and cleared her airway. The officer credited structured first aid training for giving him the confidence and technique to act. Experts from the American Red Cross noted that choking is one of the leading causes of death in infants, yet a combination of back blows and abdominal thrusts can be life-saving in seconds. This is exactly the gap MediAssist closes: structured, step-by-step guidance for bystanders who may never have trained.

Ranchi Traffic Police Trained in CPR & First Aid - April 2025

Source: Times of India / Curesta Health Initiative, April 7, 2025

In Ranchi, Jharkhand, Curesta Health launched a landmark initiative to train Ranchi Traffic Police personnel in CPR, basic first aid, and emergency response. Traffic police are frequently the first on scene at road accidents and cardiac events. The training covered CPR techniques, airway management, and trauma stabilization. This real-world programme underscores how knowledge gaps among first responders cost lives daily and validates the need for on-demand tools like MediAssist for untrained civilians.

Hyderabad RTA: Driving Test Candidates Become First Aid Trainees

Source: The Hindu, Hyderabad Edition

Hyderabad's Regional Transport Authority (RTA) integrated first aid and CPR training into the driving test process, turning driving licence candidates into potential first responders. The initiative reflects a growing recognition that bystanders not just medical professionals save lives. Every person who learns the Heimlich manoeuvre, CPR, or basic wound care becomes a node in a life-saving network. MediAssist extends this network to everyone, everywhere, without requiring prior training.

8. How We Built It

Frontend	React Native + Next.js · PWA for offline
AI Engine	Ollama with RAG pipeline
Vector DB	Pinecone / ChromaDB for semantic search
Voice	Whisper (speech-to-text) + TTS for narration
Backend	FastAPI + Redis + Docker
Knowledge	Red Cross, AHA, St. John, WHO handbooks indexed

Safety First - No Hallucinations

- AI only answers using retrieved book passages
- Confidence threshold: low-confidence → 'Call EMS immediately' fallback
- No dosages, no drug names, no surgical advice
- Persistent disclaimer + emergency number on every screen

9. Business Model

Emergency guidance is always free. That's non-negotiable. Revenue comes from institutional and preparedness use cases.

Revenue Stream	Model	Who Pays
Free + Freemium	Core guidance always free; premium features paid	General public
Enterprise License	Annual SaaS for companies	HR & safety teams
Government B2G	National health deployments	Health ministries
API Access	Pay-per-call for integrations	Device makers, IoT
Training Courses	Paid preparedness certification	Individuals, schools

11. Why MediAssist Wins

- RAG over books not the internet. Every step is verifiable and certified.
- Offline-first works without signal in disasters, rural areas, remote locations.
- Emergency UX designed for shaking hands and blurred vision. Big text, one-tap flows.
- Adaptive asks the right questions, like a real paramedic. Static apps can't do this.
- Transparent every response cites its source passage. Trust through accountability.

Sources

All statistics and emergency protocols are drawn from:

- WHO Global Health Estimates 2023
- American Heart Association CPR Guidelines 2022 & 2025
- British Red Cross First Aid Manual, 11th Edition
- CDC Emergency Preparedness Reports
- St. John Ambulance First Aid for Life Handbook
- NIH Diabetic Emergency Incidence Data
- OSHA Workplace Injury Response Effectiveness Data
- Lancet : The Global Burden of Snakebite (Kasturiratne et al.)
- NEJM Bystander CPR Effectiveness Meta-Analysis 2022
- KSLA / WABC News : Off-duty officer saves choking toddler, Dec 28, 2025
- Curesta Health : BLS Training for Ranchi Traffic Police, Apr 7, 2025
- The Hindu : Hyderabad RTA Driving Test Candidates Trained in First Aid & CPR

MediAssist does not diagnose or replace professional medical care. Always call emergency services in a crisis.